

segregation: Segregation Analysis, Inference, and Decomposition in Python

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Summary

segregation is an open-source Python library for the measurement, analysis, inference, and decomposition of social and spatial segregation patterns. Developed as part of the PySAL ecosystem, the package provides a comprehensive framework for quantifying segregation across a wide range of demographic, socioeconomic, and geographic contexts. It supports both traditional aspatial measures and advanced spatially explicit approaches, enabling researchers to investigate how population groups are distributed across space and how these patterns evolve over time.

Segregation analysis is a central topic in urban studies, geography, sociology, demography, and public policy. The **segregation** package implements a large collection of segregation indices, including single-group, multigroup, local, and spatial measures. These methods capture different dimensions of segregation such as evenness, exposure, concentration, clustering, centralization, and spatial proximity. The package accommodates a variety of spatial representations, including adjacency relationships, distance-based interactions, and network-based structures, allowing users to select measures that align with the theoretical and empirical characteristics of their study area.

Beyond point estimation, **segregation** provides a robust inferential framework for evaluating the statistical significance of segregation measures. Users can perform single-value and comparative hypothesis tests using simulation-based approaches, facilitating rigorous assessments of whether observed segregation patterns differ from those expected under specified null models. The package also includes decomposition methods that separate differences in segregation levels into components attributable to demographic composition and spatial structure, offering deeper insight into the processes underlying segregation dynamics.

Designed to operate natively with Pandas and GeoPandas data structures, **segregation** integrates seamlessly into contemporary Python-based spatial data science workflows. As a component of the Python Spatial Analysis Library (PySAL) ecosystem (S. J. Rey and Anselin 2007; S. J. Rey et al. 2022), it follows shared principles of interoperability, transparency, reproducibility, and methodological rigor. By providing accessible, well-documented, and extensible implementations of state-of-the-art segregation measures, **segregation** serves both applied researchers investigating social inequality and methodological developers advancing quantitative approaches to segregation analysis.

Statement of need

Residential segregation, inequality, and spatial separation of population groups remain central concerns in urban studies, sociology, demography, geography, and public policy. Researchers and practitioners frequently seek to quantify the extent to which social groups are unevenly distributed across neighborhoods, cities, and regions, as well as to understand how segregation patterns vary across space and time. The increasing availability of demographic and socioeconomic data has expanded opportunities for segregation analysis, but it has also highlighted the need for accessible, transparent, and reproducible computational tools capable of implementing a wide range of segregation measures and inferential procedures.

Traditional software environments for segregation analysis often provide only a limited set of indices, rely on proprietary platforms, or require researchers to implement methods manually. These limitations can hinder reproducibility, reduce methodological transparency, and create barriers for comparing results across studies. Furthermore, advances in segregation research have introduced numerous spatially explicit measures, local indicators, decomposition techniques, and simulation-based inference methods that are not consistently available within existing analytical software.

Within the Python ecosystem, support for segregation analysis was historically fragmented. While libraries such as Pandas and GeoPandas provide essential data structures for handling tabular and spatial data, they do not natively implement segregation measures or associated inferential frameworks. Consequently, researchers often relied on custom scripts, isolated software implementations, or statistical packages in other programming languages, resulting in inconsistent workflows and difficulties in reproducing analyses.

segregation addresses these challenges by providing:

- A comprehensive and well-documented API for segregation measurement and analysis
- Native integration with Pandas, GeoPandas, and the broader PySAL ecosystem
- Support for a large collection of aspatial, spatial, local, and multigroup segregation indices
- Functions for measuring and visualizing multiscalar segregation profiles
- Simulation-based inference procedures for evaluating the statistical significance of segregation measures
- Decomposition methods that facilitate the investigation of demographic and spatial sources of segregation
- A strong emphasis on transparency, reproducibility, and methodological extensibility

These capabilities make **segregation** particularly valuable for researchers and practitioners in fields such as urban studies, sociology, demography, geography, public policy, public health, and regional science, where understanding patterns of spatial inequality and social separation is a fundamental analytical objective.

State of the field

segregation is a component of the PySAL ecosystem, which provides a comprehensive suite of tools for spatial analysis in Python. Within this ecosystem, the packages are divided in four types (libraries examples in parenthesis):

- **Lib:** Core spatial data structures, file IO. Construction and interactive editing of spatial weights matrices & graphs. Alpha shapes, spatial indices, and spatial-topological relationships (**libpysal**)
- **Explore:** Modules to conduct exploratory analysis of spatial and spatio-temporal data (**esda**, **giddy**, **inequality**, etc.)
- **Model:** Estimation of spatial relationships in data with a variety of linear, generalized-linear, generalized-additive, and nonlinear models (**spreg**, **spopt**, **tobler**, **spglm**, etc.)
- **Viz:** Visualize patterns in spatial data to detect clusters, outliers, and hot-spots (**mapclassify**, **splot**, etc.)

segregation is present in the *Explore* set of libraries and addresses the specific problem of residential segregation.

Compared to desktop GIS platforms, **segregation** offers several advantages:

- **Reproducibility:** Workflows can be scripted and version-controlled
- **Transparency:** Methods and assumptions are explicit and inspectable
- **Extensibility:** Users can modify or extend algorithms for research purposes
- **Integration:** Interpolation can be embedded within larger data science pipelines, including machine learning and statistical modeling

Segregation phenomenon is also possible to be assessed with other softwares like the R packages `OasisR` (Tivadar 2019) or `seg` (Hong 2011). However, PySAL’s `segregation` module distinguishes itself in several critical ways: supports over 40 distinct segregation indices out-of-the-box, it allows simultaneous computation of multiple indices across multiscalar geographical frameworks with fewer software dependencies, it integrates spatial networks using topological relationships (e.g., spaghetti). This allows users to measure social separation based on real-world street networks rather than assuming straight-line (Euclidean), it also has a set of function for evaluating statistical significance using simulations, and it also allows users to decompose segregation scores—identifying exactly how much segregation occurs due to spatial context.

In addition, `segregation` provides a native solution for Python users, aligning with the growing adoption of Python in geospatial and data science communities.

Software design

Originally, Cortes et al. (2020) presented `segregation` but since then many new implementations were developed and the API suffered a major revision.

‘spatially-explicit’ and ‘spatially-implicit’ follows Reardon and O’Sullivan (2004)

`tobler` is designed with attention to both computational efficiency and usability. Spatial overlay operations, which are central to areal interpolation, can be computationally intensive for large datasets. The package leverages vectorized operations and efficient geometric libraries (via GeoPandas and Shapely) to handle these tasks.

The API design emphasizes clarity and consistency, with function signatures that explicitly distinguish between extensive and intensive variables. This reduces the likelihood of common errors in interpolation workflows and encourages best practices in spatial data handling.

Additionally, `tobler` is developed with testing and documentation standards consistent with the Scientific Python ecosystem, ensuring reliability and maintainability.

Core Functionality

`tobler` organizes its functionality around several key interpolation paradigms, each corresponding to different assumptions about how variables are distributed within source zones.

Area-weighted interpolation

Area-weighted interpolation is the most basic and widely used method for transferring data between polygon layers. It assumes that variables are uniformly distributed within each source zone and allocates values to target zones in proportion to the area of overlap.

`tobler` provides efficient implementations for both **extensive variables** (e.g., population counts) and **intensive variables** (e.g., rates or densities), ensuring appropriate handling of each type (Goodchild and Lam 1980). The library also supports pycnophylactic adjustments to preserve totals where required (Tobler 1979).

Dasymetric interpolation

Dasymetric interpolation refines area-weighted approaches by incorporating ancillary data—such as land use, land cover, or remotely sensed information—to model the internal heterogeneity of source zones. For example, population may be redistributed only to residential areas rather than uniformly across all land (Mennis and Hultgren 2006; Eicher and Brewer 2001; Reibel and Agrawal 2007).

tobler supports both vector- and raster-based dasymetric workflows, allowing users to integrate a wide range of auxiliary datasets. This is particularly useful in urban and environmental applications where fine-scale heterogeneity is important.

Model-based interpolation

Beyond deterministic approaches, **tobler** includes model-based methods that use statistical or machine learning techniques to estimate spatial distributions. These approaches can incorporate covariates and capture more complex spatial patterns, providing improved accuracy in many contexts (Flowerdew and Green 1992; Flowerdew and Aitkin 1982).

The design of **tobler** allows these methods to be extended and customized, making the package a useful platform for methodological research in spatial interpolation.

Integration with GeoPandas

All core functions in **tobler** operate directly on GeoPandas GeoDataFrames, minimizing friction in typical workflows. Users can pass source and target datasets as GeoDataFrames, specify variables of interest, and obtain interpolated results as new GeoDataFrames. This design leverages the broader geospatial Python stack, including Shapely for geometry operations and pandas for tabular data handling.

Example workflow

A typical area-weighted interpolation in **tobler** can be implemented as follows:

```
from tobler.area_weighted import area_interpolate

result = area_interpolate(
    source_df,
    target_df,
    extensive_variables=["population"],
    intensive_variables=["income"]
)
```

This operation transfers population counts and income measures from the source geometries to the target geometries, handling each variable type (extensive/intensive) appropriately.

When additional information about within-zone heterogeneity is available, dasymetric interpolation can be used to refine estimates. For example, population counts may be redistributed using a land cover raster to exclude uninhabited areas:

```
from tobler.dasymetric import masked_area_interpolate

result = masked_area_interpolate(
    raster="raster_file_name.tif",
    source_df,
    target_df,
    pixel_values = [21,22,23,24],
    extensive_variables=["population"]
)
```

This approach assumes the user have a raster data of his own that can be read by rasterio¹. In this example, **tobler** allows a flexible approach where the user can pass which pixels are to be assumed inhabited through **pixel_values** resulting in a more realistic spatial distribution. Similarly, the user can execute a model-based approach using the **tobler.model.glm** function.

Figure 1 illustrates an example comparing interpolated values derived from different spatial configurations, highlighting how results may vary depending on the underlying geometry and interpolation approach.

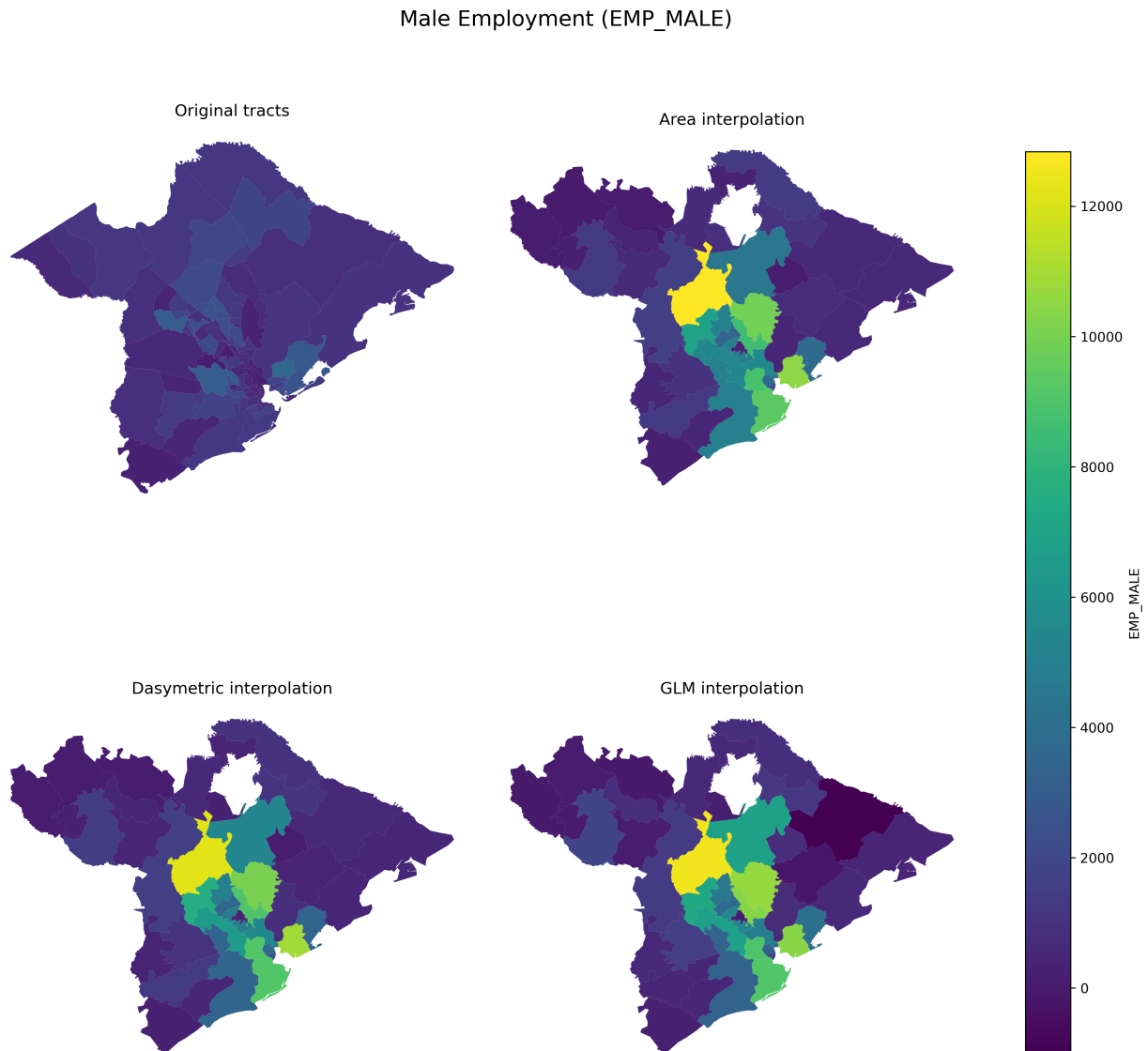


Figure 1: Example of **tobler** usage for an extensive variable (male employment population) in Charleston, SC, comparing census tracts and ZCTAs.

¹A common example is the ones available at the National Land Cover Database.

Research impact statement

The package is actively used by the research community to transfer the data between various types of geographic boundaries. This is not limited to specific applications but covers use cases from continental analysis of emissions and health (La Porta and Zapperi 2024a), analysis of urban form and function (Fleischmann and Arribas-Bel 2022), redistribution of census data to school districts for assessment of the Clean School Bus Rebate Program (Osia et al. 2025a), quantification of radon exposure (Lee et al. 2026a), or harmonization of vector and raster data for computer vision tasks (Fleischmann and Arribas-Bel 2024).

Moreover, the package is relied on in downstream software as **atlasbr** for harmonization of Brazilian urban data (Oliveira and Paiva Neto 2025), and is referred to in the **pygridmap** package by Eurostat (Grazzini and Gaffuri 2020) as a reference implementation. The **tobler** package has made tangible contributions to spatial science, pedagogy, and applications in government and industry. In academia, the package is used as part of a data-processing pipeline for research that examines the spatial-contextual influence on a variety of outcomes, including segregation (Wei et al. 2022), housing policy (S. Rey and Knaap 2022), education policy (S. J. Rey et al. 2024; Osia et al. 2025b), and pollution exposure (Lee et al. 2026b; La Porta and Zapperi 2024b). It is also used in environmental science (Hu et al. 2023) and regionalization research (Feng, Rey, and Wei 2022).

In spatial data science education, **tobler** has become an integral part of many many curricula. It is included in popular pedagogical resources including two textbooks (S. J. Rey, Arribas-Bel, and Wolf 2023; Knaap 2026), and is taught in graduate and undergraduate courses in universities across the globe, including the University of California (Berkeley, Irvine, and Riverside campuses), San Diego State University, Charles University, University of Liverpool, Bristol University, the University of Chicago, Northern Arizona University, and Temple University.

I took some liberty with a couple of these...we might want to check with Luc and Levi

In the public sector, the **tobler** package is used as part of a processing pipeline that powers urban planning and policymaking, including two highly visible projects from the Turing Institute, DemoLand and UrbanGrammar. **Martin/Dani could you confirm and add a sentence or two??**

AI usage disclosure

No generative AI or LLMs were used for code production for **tobler** or the writing of this paper.

Acknowledgements

tobler is developed as part of the PySAL community, which brings together researchers and developers working on spatial analysis methods and software. The project builds on decades of research in areal interpolation, dasymetric mapping, and spatial data science, and benefits from contributions across the open-source geospatial community.

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